

CHAPTER – XII

APPENDIX

12.1 – Source Code

app.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/dashboard01')
def dashboard01():
    return render_template('dashboard01.html')

@app.route('/dashboard02')
def dashboard02():
    return render_template('dashboard02.html')

@app.route('/dashboard03')
def dashboard03():
    return render_template('dashboard03.html')

@app.route('/dashboard04')
def dashboard04():
    return render_template('dashboard04.html')

@app.route('/story')
def story():
    return render_template('story.html')

if __name__ == '__main__':
    app.run(debug=True)
```

navbar.html

```
nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
```

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```
<a class="navbar-brand" href="/">Electricity Analysis</a>

<button class="navbar-toggler"
  data-bs-toggle="collapse"
  data-bs-target="#menu">
  <span class="navbar-toggler-icon"></span>
</button>

<div class="collapse navbar-collapse" id="menu">

  <ul class="navbar-nav ms-auto">

    <li class="nav-item">
      <a class="nav-link" href="/">Home</a>
    </li>

    <li class="nav-item">
      <a class="nav-link" href="/about">About</a>
    </li>

    <li class="nav-item">
      <a class="nav-link"
href="/dashboard01">Dashboard 1</a>
    </li>

    <li class="nav-item">
      <a class="nav-link"
href="/dashboard02">Dashboard 2</a>
    </li>

    <li class="nav-item">
      <a class="nav-link"
href="/dashboard03">Dashboard 3</a>
    </li>

    <li class="nav-item">
      <a class="nav-link"
href="/dashboard04">Dashboard 4</a>
    </li>

    <li class="nav-item">
      <a class="nav-link" href="/story">Story</a>
    </li>

  </ul>
```

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```
</div>
```

```
</div>
</nav>
```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>Home</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootst
rap.min.css" rel="stylesheet">

<link rel="stylesheet"
href="{{url_for('static',filename='css/style.css')}}">

</head>
<body>

<!-- Navbar -->
<nav class="navbar navbar-expand-lg custom-navbar">
<div class="container">

<a class="navbar-brand text-white fw-bold" href="/">
⚡ Electricity Analysis
</a>

<button class="navbar-toggler bg-light"
data-bs-toggle="collapse"
data-bs-target="#menu">

<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse" id="menu">

<ul class="navbar-nav ms-auto">

<li class="nav-item">
<a class="nav-link active" href="/">Home</a>
```

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```
</li>

<li class="nav-item">
<a class="nav-link" href="/about">About</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/dashboard01">Dashboard 1</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/dashboard02">Dashboard 2</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/dashboard03">Dashboard 3</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/dashboard04">Dashboard 4</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/story">Story</a>
</li>

</ul>

</div>
</div>
</nav>
```

```
<section class="hero">
<div class="container text-center">
<h1>Plugging into the Future</h1>
<h3>
An Exploration of Electricity Consumption Patterns using Tableau
</h3>

<br>

<a href="/dashboard01" class="btn btn-warning btn-lg">
Explore Dashboard
</a>
```

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```
</div>

</section>
```

```
</body>
</html>
</html>
```

about.html

```
<!DOCTYPE html>
<html>
<head>

<title>About</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootst
trap.min.css" rel="stylesheet">

</head>

<body>

<div class="container mt-5">

<h1 class="text-primary text-center">
About Project
</h1>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">
<title>Project Overview & Team Section</title>

<style>
    *{
        margin:0;
        padding:0;
        box-sizing:border-box;
        font-family: Arial, sans-serif;
    }

    body{
        background:#f4f6f9;
        padding:40px 20px;
```

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```
}

.container{
  max-width:1200px;
  margin:auto;
}

/* Project Overview */
.project-overview{
  background:#fff;
  padding:30px;
  border-radius:16px;
  box-shadow:0 4px 12px rgba(0,0,0,0.08);
  margin-bottom:30px;
}

.project-overview h2{
  color:#1e293b;
  margin-bottom:15px;
}

.project-overview p{
  color:#555;
  line-height:1.7;
}

/* Team Section */
.team-section h2{
  margin-bottom:20px;
  color:#1e293b;
}

.team-grid{
  display:grid;
  grid-template-columns:repeat(3, 1fr);
  gap:20px;
}

.team-card{
  background:#fff;
  padding:25px;
  border-radius:16px;
  text-align:center;
  box-shadow:0 4px 12px rgba(0,0,0,0.08);
  transition:0.3s ease;
}

.team-card:hover{
```

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```
        transform:translateY(-5px);
    }

    .team-card img{
        width:100px;
        height:100px;
        border-radius:50%;
        object-fit:cover;
        margin-bottom:15px;
        border:4px solid #e2e8f0;
    }

    .team-card h3{
        color:#1e293b;
        margin-bottom:5px;
    }

    .role{
        color:#3b82f6;
        font-weight:600;
        margin-bottom:12px;
    }

    .team-card p{
        color:#666;
        font-size:14px;
        line-height:1.6;
    }

    @media (max-width: 992px){
        .team-grid{
            grid-template-columns:repeat(2, 1fr);
        }
    }

    @media (max-width: 768px){
        .team-grid{
            grid-template-columns:1fr;
        }
    }
</style>
</head>
<body>

<div class="container">

    <!-- Project Overview -->
    <section class="project-overview">
```

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```
<h2>Project Overview</h2>
<p>
    This project aims to build a modern web application
that streamlines
    business operations and enhances team collaboration.
The solution
    includes user management, reporting dashboards,
workflow automation,
    and real-time communication features.
</p>
</section>

<!-- Team Details -->
<section class="team-section">
    <h2>Our Team</h2>

    <div class="team-grid">

        <div class="team-card">
            
            <h3>Burri Nikshitha</h3>
            <div class="role">Team Leader</div>
            <p>
                Leads project planning, stakeholder
communication,
                and delivery management.
            </p>
        </div>

        <div class="team-card">
            
            <h3>Selvaraj Yaswini</h3>
            <div class="role">Team Member</div>
            <p>
                Creates intuitive user experiences and modern
interface
                designs for the application.
            </p>
        </div>

        <div class="team-card">
            
            <h3>Samadhi Harshitha</h3>
            <div class="role">Team Member</div>
            <p>
```


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Develops scalable application features and maintains backend and frontend systems.

```
</p>
</div>
```

```
</div>
</section>
```

```
</div>
```

```
</body>
</html>
```

```
</section>
</div>
```

```
</body>
</html>
```

dashboard01.html

```
<!DOCTYPE html>
<html>
<head>

<title>Dashboard 1</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
```

```
</head>
```

```
<body>
```

```
<div class="container mt-5">
```

```
<h1 class="text-success text-center">
Dashboard 1
</h1>
```

```
<hr>
```

```
<h3>
State Consumption Analysis
</h3>
```

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```
<div class="card shadow">

<div class="card-body">

<!-- Tableau Embed -->

    <div class='tableauPlaceholder' id='viz1781347644746'
style='position: relative'><noscript><a href='#'><img
alt='Dashboard 1 '
src='https://public.tableau.com/static/images/GZ/GZ2SXKWKH/1_rss.png' style='border: none'
/></a></noscript><object class='tableauViz'
style='display:none;'><param name='host_url'
value='https%3A%2F%2Fpublic.tableau.com%2F' /> <param
name='embed_code_version' value='3' /> <param name='path'
value='shared/GZ/GZ2SXKWKH' /> <param name='toolbar' value='yes'
/><param name='static_image'
value='https://public.tableau.com/static/images/GZ/GZ2SXKWKH/1.png' /> <param name='animate_transition'
value='yes' /><param name='display_static_image' value='yes'
/><param name='display_spinner' value='yes' /><param
name='display_overlay' value='yes' /><param name='display_count'
value='yes' /><param name='language' value='en-US' /><param
name='filter' value='publish=yes' /></object></div>
<script type='text/javascript'>                var divElement
= document.getElementById('viz1781347644746');
var vizElement = divElement.getElementsByTagName('object')[0];
if ( divElement.offsetWidth > 800 ) {
vizElement.style.width='1200px';vizElement.style.height='827px';}
else if ( divElement.offsetWidth > 500 ) {
vizElement.style.width='1200px';vizElement.style.height='827px';}
else {
vizElement.style.width='100%';vizElement.style.height='1227px';}
var scriptElement = document.createElement('script');
scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>

    src="YOUR_TABLEAU_LINK"
    width="100%"
    height="600">

</div>
```

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```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

dashboard02.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Dashboard 2</title>
```

```
<link
```

```
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
```

```
</head>
```

```
<body>
```

```
<div class="container mt-5">
```

```
<h1 class="text-primary text-center">
```

```
Dashboard 2
```

```
</h1>
```

```
<hr>
```

```
<h3>
```

```
Regional Electricity Consumption
```

```
</h3>
```

```
<div class="card shadow">
```

```
<div class="card-body">
```

```
<div class="tableau-container">
```

```
<div class="tableauPlaceholder" id="viz1781347799814">
```

```
<object class="tableauViz">
```

```
<param name="host_url"
```

```
value="https%3A%2F%2Fpublic.tableau.com%2F" />
```

```
<param name="embed_code_version" value="3" />
```

```
<param name="site_root" value="" />
```

```
<param name="name"
```

```
value="PluggingintotheFutureAnExplorationofElectricityConsumptionP  
atterns-Dashboard-02/Dashboard2" />
```

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```
        <param name="tabs" value="no" />
        <param name="toolbar" value="yes" />
    </object>
</div>
</div>

<style>
.tableau-container {
    width: 100%;
    max-width: 1200px;
    margin: 0 auto;
}

.tableauPlaceholder,
.tableauViz {
    width: 100% !important;
}

.tableauViz {
    min-height: 700px;
}
</style>

<script>
var divElement = document.getElementById('viz1781347799814');
var vizElement = divElement.getElementsByTagName('object')[0];

vizElement.style.width = '100%';
vizElement.style.height = '800px';

var scriptElement = document.createElement('script');
scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>
src="YOUR_TABLEAU_LINK"
width="100%"
height="600">

</div>

</div>

</div>

</body>
</html>
```

Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

dashboard03.html

```
<!DOCTYPE html>
<html>
<head>

<title>Dashboard 3</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">

</head>

<body>

<div class="container mt-5">

<h1 class="text-danger text-center">
Dashboard 3
</h1>

<hr>

<h3>
Metro City Usage Analysis
</h3>

<div class="card shadow">

<div class="card-body">

<div class='tableauPlaceholder' id='viz1781347908217'
style='position: relative'><noscript><a href='#'><img
alt='Dashboard 3 '
src='https://public.tableau.com/static/images/Pl/PluggingintotheFutureAnExplorationofElectricityConsumptionPatterns-Dashboard-03/Dashboard3/1_rss.png' style='border:
none' /></a></noscript><object class='tableauViz'
style='display:none;'><param name='host_url'
value='https://public.tableau.com/' /> <param
name='embed_code_version' value='3' /> <param name='site_root'
value='' /><param name='name'
value='PluggingintotheFutureAnExplorationofElectricityConsumptionPatterns-Dashboard-03/Dashboard3' /><param name='tabs'
value='no' /><param name='toolbar' value='yes' /><param
name='static_image'
```

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```
value='https://public.tableau.com/static/images/7;PluggingintotheFutureAnExplorationofElectricityConsumptionPatterns-Dashboard-03;Dashboard3;1.png' /> <param
name='animate_transition' value='yes' /><param
name='display_static_image' value='yes' /><param
name='display_spinner' value='yes' /><param name='display_overlay'
value='yes' /><param name='display_count' value='yes' /><param
name='language' value='en-US' /><param name='filter'
value='publish=yes' /></object></div>                <script
type='text/javascript'>                var divElement =
document.getElementById('viz1781347908217');
var vizElement = divElement.getElementsByTagName('object')[0];
if ( divElement.offsetWidth > 800 ) {
vizElement.style.width='1000px';vizElement.style.height='827px';}
else if ( divElement.offsetWidth > 500 ) {
vizElement.style.width='1000px';vizElement.style.height='827px';}
else {
vizElement.style.width='100%';vizElement.style.height='927px';}
var scriptElement = document.createElement('script');
scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>
```

```
src="YOUR_TABLEAU_LINK"
width="100%"
height="600">
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

dashboard04.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>About</title>
```

```
<link
```

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```
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
```

```
</head>
```

```
<body>
```

```
<div class="container mt-5">
```

```
<h1 class="text-primary text-center">
```

```
About Project
```

```
</h1>
```

```
<!DOCTYPE html>
```

```
<html lang="en"><head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Project Overview & Team Section</title>
```

```
<style>
```

```
  *{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family: Arial, sans-serif;
  }
```

```
  body{
    background:#f4f6f9;
    padding:40px 20px;
  }
```

```
  .container{
    max-width:1200px;
    margin:auto;
  }
```

```
  /* Project Overview */
  .project-overview{
    background:#fff;
    padding:30px;
    border-radius:16px;
    box-shadow:0 4px 12px rgba(0,0,0,0.08);
    margin-bottom:30px;
  }
```

```
  .project-overview h2{
    color:#1e293b;
```

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```
margin-bottom:15px;
}

.project-overview p{
  color:#555;
  line-height:1.7;
}

/* Team Section */
.team-section h2{
  margin-bottom:20px;
  color:#1e293b;
}

.team-grid{
  display:grid;
  grid-template-columns:repeat(3, 1fr);
  gap:20px;
}

.team-card{
  background:#fff;
  padding:25px;
  border-radius:16px;
  text-align:center;
  box-shadow:0 4px 12px rgba(0,0,0,0.08);
  transition:0.3s ease;
}

.team-card:hover{
  transform:translateY(-5px);
}

.team-card img{
  width:100px;
  height:100px;
  border-radius:50%;
  object-fit:cover;
  margin-bottom:15px;
  border:4px solid #e2e8f0;
}

.team-card h3{
  color:#1e293b;
  margin-bottom:5px;
}

.role{
```


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```
    color:#3b82f6;
    font-weight:600;
    margin-bottom:12px;
}

.team-card p{
    color:#666;
    font-size:14px;
    line-height:1.6;
}

@media (max-width: 992px){
    .team-grid{
        grid-template-columns:repeat(2, 1fr);
    }
}

@media (max-width: 768px){
    .team-grid{
        grid-template-columns:1fr;
    }
}
</style>
</head>
<body>

<div class="container">

    <!-- Project Overview -->
    <section class="project-overview">
        <h2>Project Overview</h2>
        <p>
            This project aims to build a modern web application
            that streamlines
            business operations and enhances team collaboration.
            The solution
            includes user management, reporting dashboards,
            workflow automation,
            and real-time communication features.
        </p>
    </section>

    <!-- Team Details -->
    <section class="team-section">
        <h2>Our Team</h2>

        <div class="team-grid">
```



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```
<div class="team-card">
  
  <h3>Burri Nikshitha</h3>
  <div class="role">Team Leader</div>
  <p>
    Leads project planning, stakeholder
communication,
    and delivery management.
  </p>
</div>

<div class="team-card">
  
  <h3>Selvaraj Yaswini</h3>
  <div class="role">Team Member</div>
  <p>
    Creates intuitive user experiences and modern
interface
    designs for the application.
  </p>
</div>

<div class="team-card">
  
  <h3>Samadhi Harshitha</h3>
  <div class="role">Team Member</div>
  <p>
    Develops scalable application features and
maintains
    backend and frontend systems.
  </p>
</div>
</div>
</section>
</div>

</body>
</html>

</section>
</div>

</body>
```



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```
</html>
```

story.html

```
<!DOCTYPE html>
<html>
<head>

<title>Story</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">

</head>

<body>

<div class="container mt-5">

<h1 class="text-info text-center">
Story
</h1>

<hr>

<div class="card shadow">

<div class="card-body">

<p>
The Tableau story presents a complete view of electricity
consumption patterns before and after lockdown, regional
usage, quarterly analysis, metro city trends and annual
consumption growth.
</p>

<div class='tableauPlaceholder' id='viz1781348269812'
style='position: relative'><noscript><a href='#'><img alt='Story 1
',
src='https://public.tableau.com/static/images/PluggingintotheFutureAnExplorationofElectricityConsumptionP
atterns-Story1_rss.png' style='border: none'
/></a></noscript><object class='tableauViz'
style='display:none;'><param name='host_url'
value='https://public.tableau.com/' /> <param
name='embed_code_version' value='3' /> <param name='site_root'
```

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```
value='' /><param name='name'
value='PluggingintotheFutureAnExplorationofElectricityConsumptionP
atterns-Story&#47;Story1' /><param name='tabs' value='no' /><param
name='toolbar' value='yes' /><param name='static_image'
value='https:&#47;&#47;public.tableau.com&#47;static&#47;images&#4
7;Pl&#47;PluggingintotheFutureAnExplorationofElectricityConsumptio
nPatterns-Story&#47;Story1&#47;1.png' /> <param
name='animate_transition' value='yes' /><param
name='display_static_image' value='yes' /><param
name='display_spinner' value='yes' /><param name='display_overlay'
value='yes' /><param name='display_count' value='yes' /><param
name='language' value='en-US' /><param name='filter'
value='publish=yes' /></object></div>                <script
type='text/javascript'>                var divElement =
document.getElementById('viz1781348269812');
var vizElement = divElement.getElementsByTagName('object')[0];
vizElement.style.width='1016px';vizElement.style.height='991px';
var scriptElement = document.createElement('script');
scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>
src="YOUR_TABLEAU_STORY_LINK"
width="100%"
height="600">
```

</div>

</div>

</div>

</body>

</html>

style.css

```
body{
background:#f4f6f9;
font-family:Segoe UI;
}

.custom-navbar{
background:linear-gradient(90deg,#ff512f,#dd2476,#6a11cb,#2575fc);
}

.nav-link{
```

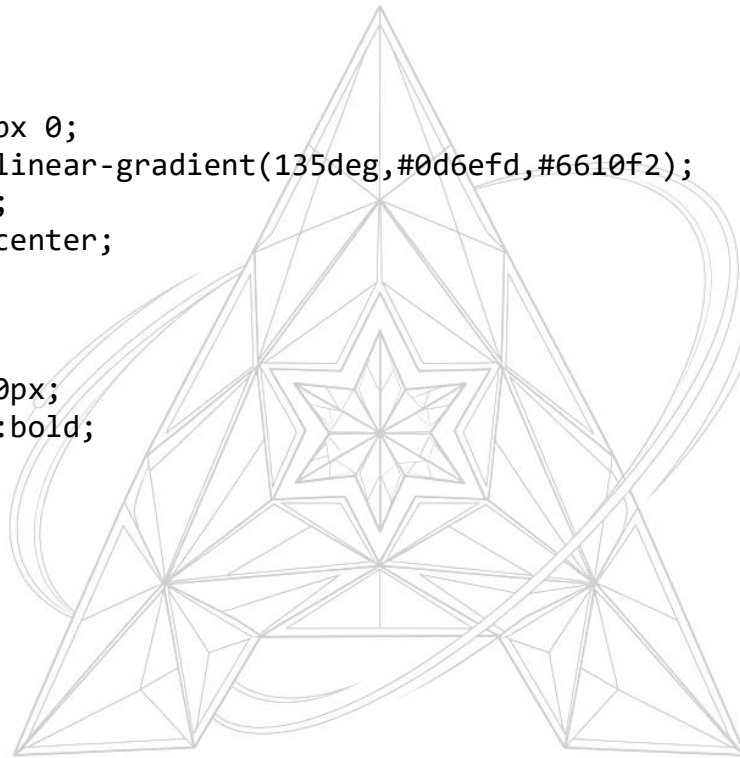
Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

```
color:white !important;  
font-weight:600;  
margin-left:10px;  
}
```

```
.nav-link:hover{  
background:white;  
color:black !important;  
border-radius:20px;  
}
```

```
.hero{  
padding:180px 0;  
background:linear-gradient(135deg,#0d6efd,#6610f2);  
color:white;  
text-align:center;  
}
```

```
.hero h1{  
font-size:60px;  
font-weight:bold;  
}
```



ALPHA
— INNOVATORS —
—

